

Customer Sales & Shopping Behavior Analysis

1. Project Overview :

Customer Sales & Shopping Behavior Analysis is a data analytics project that aims to understand customer purchasing patterns, sales trends, and shopping preferences using transactional data. The project analyzes customer demographics, product categories, purchase amounts, discounts, subscriptions, and shopping trends to identify meaningful business insights. The analysis was performed using **Python** for data cleaning and exploratory data analysis (EDA), **PostgreSQL** for business analysis, and **Power BI** for creating an interactive dashboard. The insights from this project help businesses improve customer engagement, optimize marketing strategies, and make better business decisions to increase sales and customer satisfaction.

2. Dataset Summary :

The dataset used in this project contains customer sales and shopping information from a retail business. It includes **3,900 records** and **18 columns**, covering customer demographics, purchase details, and shopping behavior. The dataset provides valuable insights into customer preferences, purchase patterns, payment methods, discounts, review ratings, and seasonal shopping trends. It was used to perform data analysis, answer business questions, and create an interactive Power BI dashboard.

3. Dataset Summary :

- **Rows** : 3,900
- **Columns** : 18
- **Key Features** :
 - Customer Demographics : Age, Gender, Location, Subscription Status
 - Purchase Details : Item Purchased, Category, Purchase Amount, Season, Size, Color
 - Shopping Behavior : Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type

- **Missing Values :** The dataset contained **37 missing values** in the **Review Rating** column, which were handled during the data cleaning process.

4. Tools & Technologies Used :

Tool	Purpose
Python	Data cleaning, preprocessing, feature engineering, and exploratory data analysis (EDA)
Pandas	Data manipulation and analysis
PostgreSQL	Data analysis using SQL queries
Power BI	Creating an interactive dashboard and data visualization
Jupyter Notebook	Writing and running Python code

5. Data Preparation & Exploratory Analysis (Python) :

- **Data Loading :**

- Loaded the dataset into Jupyter Notebook using pandas.
- Checked the dataset structure and verified the number of rows and columns.

- **Data Understanding :**

- Examined the column names and data types. Used df.info() to check structure and .describe() for summary statistics.
- Reviewed the dataset to understand the available customer and sales information.

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                                Non-Null Count  Dtype  
---  -
 0   Customer ID                          3900 non-null   int64   
 1   Age                                   3900 non-null   int64   
 2   Gender                               3900 non-null   object  
 3   Item Purchased                       3900 non-null   object  
 4   Category                             3900 non-null   object  
 5   Purchase Amount (USD)                3900 non-null   int64   
 6   Location                             3900 non-null   object  
 7   Size                                  3900 non-null   object  
 8   Color                                3900 non-null   object  
 9   Season                               3900 non-null   object  
10  Review Rating                        3863 non-null   float64  
11  Subscription Status                 3900 non-null   object  
12  Shipping Type                       3900 non-null   object  
13  Discount Applied                    3900 non-null   object  
14  Promo Code Used                     3900 non-null   object  
15  Previous Purchases                  3900 non-null   int64   
16  Payment Method                      3900 non-null   object  
17  Frequency of Purchases               3900 non-null   object  
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

```
# For both Numerical and Categorical Columns
df.describe(include="all")
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

- **Data Cleaning :** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Feature Engineering :**
 - Created the **age_group** column.
 - Converted **purchase_frequency_days** into numerical values for easier analysis.
- **Data Consistency Check :** Verified if **discount_applied** and **promo_code_used** were redundant; dropped **promo_code_used**.

- **Database Integration :** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

6. Data Analysis using SQL (PostgreSQL) :

After cleaning the dataset in Python, the data was imported into PostgreSQL for business analysis. SQL queries were used to answer important business questions related to customer spending, product performance, discounts, subscriptions, and customer loyalty. This analysis helped identify customer segments, high-performing products, and factors that influence purchasing behavior.

SQL Analysis Performed :

- **Revenue by Gender :** Compared total revenue generated by male and female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

- **High-Spending Discount Users :** Identified customers who used discounts but still spent more than the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
Total rows: 839		Query complete 00:00

- Top 5 Products by Rating : Found the products with the highest average review ratings.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

- Shipping Type Comparison : Compared average purchase amounts between Standard and Express shipping.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

- Subscribers vs. Non-Subscribers : Compared average spending and total revenue of subscribed and non-subscribed customers.

	subscription_status text	total_customers bigint	avg_spent numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

- Discount-Dependent Products : Identified products with the highest percentage of discounted purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

- Customer Segmentation : Classified customers as New, Returning, or Loyal based on their purchase history.

	customer_segment text	Number_of_Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

- Top 3 Products per Category : Found the most purchased products in each product category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

- Repeat Buyers & Subscriptions : Analyzed whether customers with more than five previous purchases were more likely to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

- Revenue by Age Group : Calculated revenue contribution from different age groups.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

7. Dashboard in Power BI :

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



8. Business Recommendations :

- **Boost Subscriptions :** Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs :** Reward repeat buyers to move them into the “Loyal” segment.
- **Review Discount Policy :** Balance sales boosts with margin control.
- **Product Positioning :** Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing :** Focus efforts on high-revenue age groups and express-shipping users.